



MTE 426

Programmable Logic Controller (PLC)

Mini Project

Dairy Production Factory PLC Program

Submitted by

Mostafa Eissa 120210241

Abdelrahman Hisham 120210254

Abdullah Essam 120210180

Submitted to:

Prof. Mohamed G. Alkalla

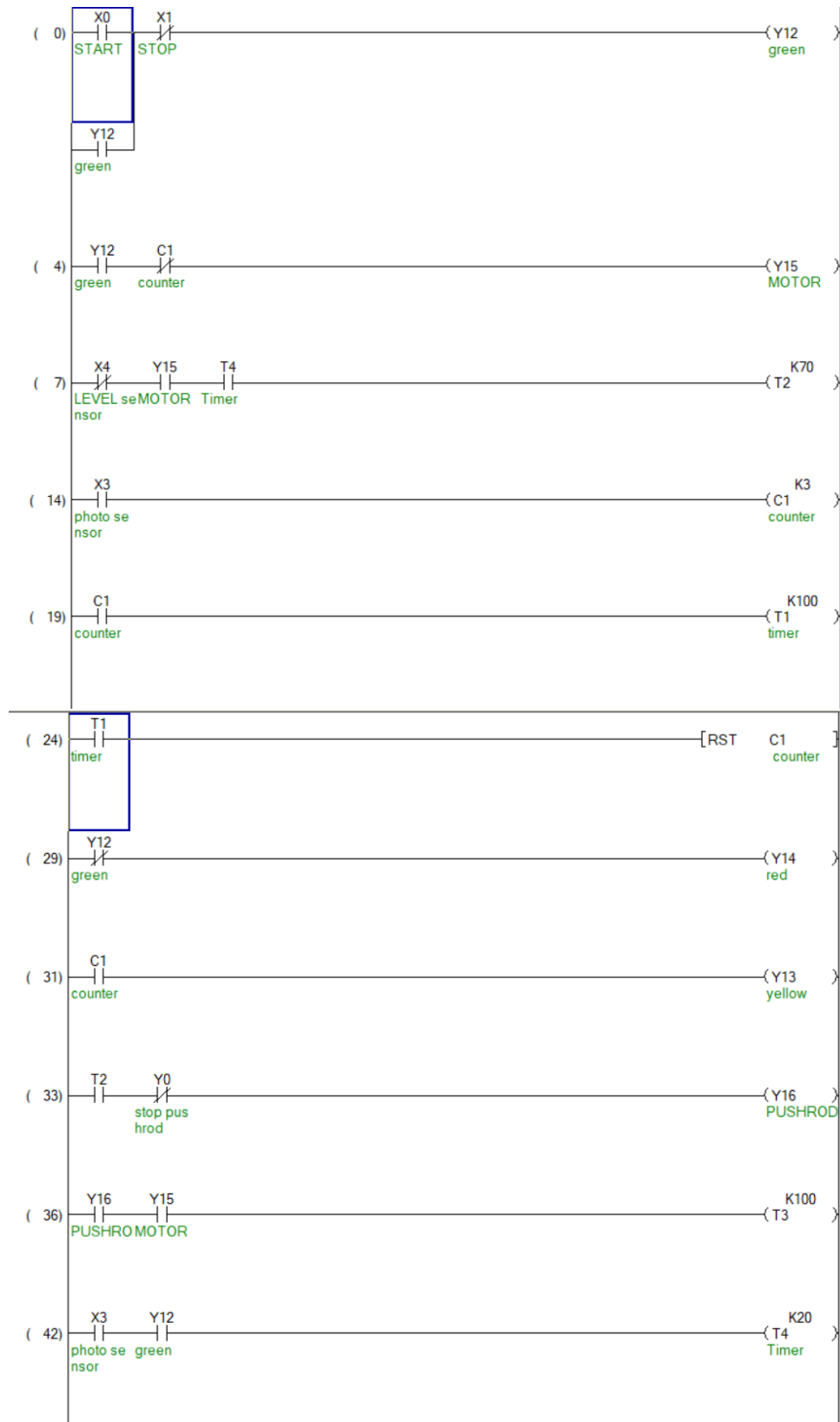
Table of Contents

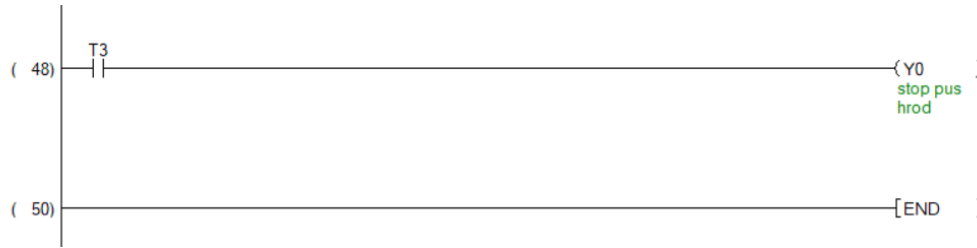
1. Laddar Diagram.....	3
2. Inputs/Outputs	4
3. Suggest a suitable sensor to detect whether the milk pack (cardboard/carton) is full or empty and another sensor for counting the packs.	4
4. The Programming Process	8

Table of Figures

Figure (1): Capacitive Proximity Switch	4
Figure (2): Photoelectric Sensor (Diffuse Type)	5

1. Ladder Diagram





2. Inputs/Outputs

	Inputs	Outputs			Timers	Counters	
I1.1	start	Q1.2	motor	T1	ON-Delay Timer (K100)	C1	Counter up (3)
I1.2	stop	Q1.3	Green	T2	ON-Delay Timer (K70)		
I1.3	Level sensor	Q1.4	Push-rod motor	T3	ON-Delay Timer (K100)		
I1.4	Photo sensor	Q1.5	Red	T4	ON-Delay Timer (K20)		
		Q1.6	Yellow				
		Q1.7	Stop push rod				

Suggest a suitable sensor to detect whether the milk pack (cardboard/carton) is full or empty and another sensor for counting the packs.

Sensor for Detecting Full or Empty Milk Packs:

A **Capacitive Proximity Sensor** is a suitable choice.

Why Capacitive Proximity Sensor?

- Capacitive sensors can detect the presence of liquids inside the carton because they sense changes in dielectric properties.
- They are non-invasive and can detect the contents without physical contact.
- Suitable for detecting liquids inside non-metallic containers (like cardboard or plastic).

Recommended Specifications:

- Sensing Distance: ~5-10 mm (depending on the carton thickness and sensor model).
- Housing Material: Plastic (for food-safe applications).
- Output: PNP or NPN (depending on your PLC's input requirements).

Placement:

- Install the sensor close to the side or bottom of the carton where the presence of liquid can be detected.



Figure (1): Capacitive Proximity Switch.

Working Principle:**1. Electric Field Generation:**

- The sensor generates an electric field around its sensing electrode.

2. Capacitance Changes with Target Presence:

- When a target object (metallic or non-metallic) comes close to the sensor:
 - The capacitance changes due to the dielectric properties of the object (e.g., liquid, cardboard, plastic).
 - For conductive objects, the object itself acts as the second capacitor plate.
 - For non-conductive objects, the material's dielectric constant alters the capacitance between the sensing electrode and ground.

3. Oscillation Amplitude Change:

- The change in capacitance causes a change in the amplitude of the oscillator's signal.

4. **Signal Detection:**

- The detector circuit identifies the change in the oscillation signal and processes it.

5. **Output Signal Switching:**

- The output circuit converts the processed signal into a binary output (ON/OFF) to indicate the presence or absence of the target.

Sensor for Counting Milk Packs:

A **Photoelectric Sensor (Diffuse Type)** is ideal for counting the milk cartons as they move on a conveyor belt.

Why Photoelectric Sensor?

- Photoelectric sensors can detect the presence of objects (like milk cartons) based on light reflection or interruption.
- They work well with various materials, including cardboard.
- High-speed response makes them perfect for counting objects on moving conveyor belts.

Recommended Specifications:

- Type: Diffuse Reflective or Through-Beam.
- Sensing Range: ~100-500 mm (adjust based on your conveyor setup).
- Output: PNP or NPN.
- Response Time: Fast enough to handle the speed of your conveyor.

Placement:

- Install the sensor facing the conveyor belt, where it can detect each milk pack as it passes by.

Working Principle:

1. **Emission of Light:**

- The sensor's emitter sends out a continuous or pulsed beam of light toward the detection zone.

2. **Reflection from Object:**

- When a milk pack (or any object) passes in front of the sensor:

- The light beam hits the surface of the milk pack.
- The object reflects the light back to the sensor's receiver.

3. **Light Detection:**

- The receiver detects the reflected light.
- If the intensity of the reflected light exceeds the sensor's threshold, it registers the object as detected.

4. **Signal Processing:**

- The sensor processes the signal from the receiver.
- The output circuit sends a signal (ON/OFF) to the PLC or counter system.

5. **Counting the Milk Packs:**

- Every time the sensor detects an object, it triggers a pulse to the PLC or counter.
- The PLC increases the count for each detected milk pack.



Figure 2: Photoelectric Sensor (Diffuse Type)

3. The Programming Process

Rung 0: Start/Stop System

- **Logic:**
 - X0 (Start) and X1 (Stop) are push buttons.
 - Y12 (Green Lamp) indicates the system is running.
 - The circuit latches Y12 ON when X0 is pressed.
 - Pressing X1 will de-latch and stop the system.
- **Purpose:**
 - This rung ensures that the system starts when the start button is pressed and stops when the stop button is pressed.



Rung 4: Counter Increment

- **Logic:**
 - Y12 ensures the system is running.
 - C1 (Counter) increments when X3 (Proximity Sensor) detects a milk pack.
- **Purpose:**
 - This rung counts the milk packs passing on the conveyor. Each detection by the sensor increments the counter by 1.



Rung 7: Motor Control with Level Sensor

- **Logic:**
 - X4 (Level Sensor) detects whether the milk carton is full or empty.
 - Y15 (Motor) is turned ON, and T4 (Timer) ensures that the motor runs for a set time (K70).
- **Purpose:**
 - Controls the motor operation and ensures it runs for a specific time based on the level sensor input.



Rung 14: Sensor and Motor Interaction

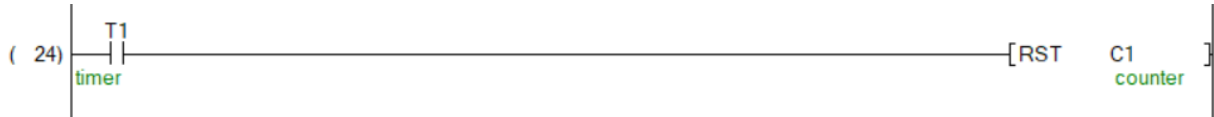
- **Logic:**
 - X3 (Sensor) detects the milk packs, interacting with the Y12 output to indicate system readiness.
- **Purpose:**
 - Verifies that the sensor is functional and interacts with the green lamp (Y12).



Rung 24 , 19 : Timer for Counter when reach the desired num of packs

- **Logic:**
 - C1 (Counter) resets after a specific count (K3).
- **Purpose:**
 - Ensures that the counter is cleared after reaching a predefined limit, preparing for the next counting cycle.





Rung 29: Indication of System not Running

- **Logic:**
 - Y14 turns ON the red lamp to indicate the system is not active.
- **Purpose:**
 - Provides visual feedback that the system is not running now



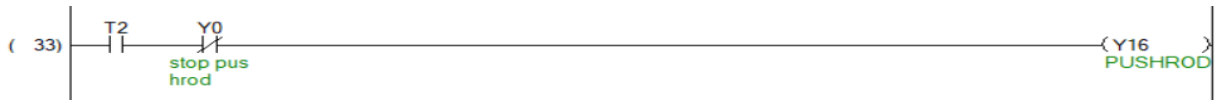
Rung 31: Yellow Lamp for process is on standby

- **Logic:**
 - C1 activates Y13 (Yellow Lamp).
- **Purpose:**
 - visual indication that process is on standby until the counter resets again..



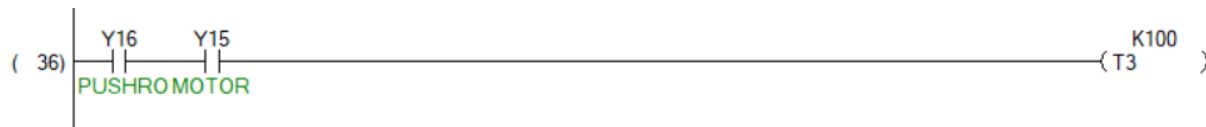
Rung 33: Push Rod Activation

- **Logic:**
 - After T2 (Timer) ended , it activates Y16 (Push Rod) to handle ejecting a pack.
- **Purpose:**
 - Handles pushing the milk pack out of the line based on timer control.



Rung 36: Push Rod and Motor Interaction

- **Logic:**
 - Y16 (Push Rod) interacts with Y15 (Motor) to make a timer for stopping the push rod later
- **Purpose:**
 - Ensures synchronization between the push rod and motor operations.



Rung 42: Sensor and timer

- **Logic:**
 - X3 (Sensor) and Y12 (Green Lamp) interact to ensure when the sensor is reading , give a timer to the timer which will be on the push rod
- **Purpose:**
 - Monitors sensor activity in conjunction with the system's operational state.



Rung 48: return of push rod

- **Logic:**
 - T3 (Timer) starts a set delay to return of push rod.
- **Purpose:**
 - Provides a delay mechanism for the push rod

